

Viewpoint Planning of Projector Placement for Spatial Augmented Reality using Star-Kernel Decomposition

Takefumi Hiraki^{*,§}
University of Tsukuba

Tomohiro Hayase^{†,§}
Fujitsu Laboratories Ltd.

Yuichi Ike^{‡,§}
Fujitsu Laboratories Ltd.

Takashi Tsuboi
Musashino University

Michio Yoshiwaki
Osaka City University

ABSTRACT

To obtain the targeted projection appearance in a spatial augmented reality system, we need to solve the projector placement problem. However, there are only heuristic solutions for viewpoint planning, which is the fundamental problem of projector placement. In this paper, we propose the star-kernel decomposition algorithm that can solve the viewpoint planning problem of an object composed of known polygons. The star-kernel decomposition algorithm decomposes a polygon into subpolygons that have non-empty star kernels. We implemented a program to obtain the placement of multiple projectors and evaluated the applicability of the obtained placement.

Index Terms: Computing methodologies—Computer graphics

1 INTRODUCTION

Spatial augmented reality (SAR), which is achieved by projecting images onto real objects, has been attractive because it allows users to experience highly realistic content without wearing devices like glasses or goggles. SAR systems often use multiple projectors to project images in order to cover a large projection area. In order to obtain the targeted appearance of projected objects in such a system, we need to optimally place multiple projectors. This problem is known as the projector planning problem [1]. The fundamental problem in projector placement planning is viewpoint planning, which locates projectors so that they can project a target object. Practically one approximates the target object by a polygon, which is a collection of faces in the 3D space and tries to find places of projectors that can see all faces. Then, the problem of viewpoint planning is related to the art gallery problem [3], which is known to be NP-hard. Hence, the problem has been tackled mainly by heuristic approaches, which need iterative optimization computation, and thus much time.

In this paper, we propose the algorithm of *star-kernel decomposition*, which is a novel viewpoint planning method of projector placement. Here, the *star-kernel* of a polygon is the set of candidate projector positions that can shed light on the faces. The star-kernel decomposition algorithm decomposes a given polygon into subpolygons that have non-empty star kernels. In addition, each resultant subpolygon is a part of the star domain's boundary, that is, a 3D region that contains a point from which the entire region is visible. Fig. 1 shows the concept of the star-kernel decomposition algorithm. This idea is inspired by existing methods in the area of image-based rendering that find the optimal camera placement for capturing the entire image of a real object of known shape. Lien [2] considers star-shaped decomposition for point set data, and Yu *et al.* [4] considers decomposition of a 3D solid shape into star domains. In contrast to the existing methods, we consider the situation where a polygon consisting of faces with normal vectors and directly search star-kernel

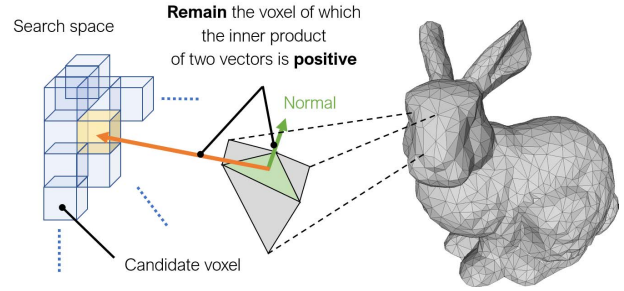


Figure 1: Conceptual figure of the proposed star-kernel decomposition algorithm.

decomposition for the polygon, which can be computed efficiently. Once one obtains the decomposition, one needs to place a projector in each star kernel. We implemented a simulation program, calculated the placement for basic objects, and verified that the obtained candidate is an appropriate projector placement.

2 METHODS

For viewpoint planning in the projector placement problem, we first introduce a star-kernel partition of a polygon in the 3D space. Assume that we are given a polygon P consisting of faces σ with a normal vector n_σ and a point $c_\sigma \in \sigma$. Practically, we consider a bounded search space $V \subseteq \mathbf{R}^3$ where we can place projectors and even set a collection of voxels of number N_v in the 3D space $\mathbf{R}^3$. In what follows, we set V to be large enough so that there exists a visible point x in V for any face σ , that is, $\langle x - c_\sigma, n_\sigma \rangle_{\mathbf{R}^3} > 0$. Now, a star-kernel partition of the polygon P is a partition $\{B_1, \dots, B_n\}$ into subpolygons of P so that

$$k_{B_m} = \bigcap_{\sigma \in B_m} \{x \in V \mid \langle x - c_\sigma, n_\sigma \rangle_{\mathbf{R}^3} > 0\} \neq \emptyset.$$

In this case, k_{B_m} is called the *star kernel* for B_m . Then we just need to place a projector in each kernel k_{B_m} .

Now, we propose the *star-kernel decomposition*, which is an iterative algorithm to find a star-kernel partition. First, we sort all faces with respect to normal vectors and center points in some manner and write $\sigma_1, \dots, \sigma_{N_f}, c_i := c_{\sigma_i}, n_i := n_{\sigma_i}$. Next, starting with the first face and the initial candidate kernel $k = V$, we shrink the candidate kernel in the following way. (i) Compute the visibility defined as follows:

$$\text{ArgVisible}(k, c_i, n_i) = \{j \mid \langle k_j - c_i, n_i \rangle_{\mathbf{R}^3} > 0\},$$

where k_j is the j -th element of k . (ii) Shrink k according to $a = \text{ArgVisible}(k, c_i, n_i)$:

$$k \leftarrow k[a] = \{k_j \mid j \in a\}.$$

We continue the above two steps for faces until $\text{ArgVisible}(k, c_i, n_i)$ becomes empty. Then the k before the one that becomes empty

^{*}e-mail: hiraki@slis.tsukuba.ac.jp

[†]e-mail: hayase.tomohiro@fujitsu.com

[‡]e-mail: ike.yuichi@fujitsu.com

[§]These authors contributed equally to this work.

Algorithm 1 Star-kernel decomposition (Python-like pseudocode)

Require: $N_v \leftarrow$ the number of voxels in the search space.
Require: args $x \leftarrow$ array of candidate voxels $\triangleright$ shape $(N_v, 3)$
Require: $N_f \leftarrow$ the number of faces
Require: args $c \leftarrow$ array of each face's center $\triangleright$ shape $(N_f, 3)$
Require: args $n \leftarrow$ array of each face's normal $\triangleright$ shape $(N_f, 3)$
Ensure: A list of kernels K
Ensure: A partition of faces π
 $K \leftarrow [], \pi \leftarrow []$
 $i \leftarrow 0$
while $i < N_f$ **do**
 $k \leftarrow x$ $\triangleright$ Initialize a candidate kernel
 $B \leftarrow []$ $\triangleright$ Initialize a partition block
 while $a \leftarrow \text{ArgVisible}(k, c[i], n[i]) \ \& \ a \neq \emptyset \ \& \ i < N_f$ **do**
 $k \leftarrow k[a]$ $\triangleright$ Shrink the candidate kernel
 Append i to B $\triangleright$ Record the index to the partition block
 $i \leftarrow i + 1$ $\triangleright$ Go to the next face
 end while
 $\triangleright$ All voxels became invisible or i reached the last face
 Append k to K and B to π $\triangleright$ Determine a kernel
end while
return K, π

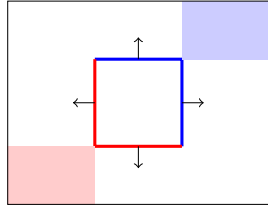


Figure 2: Example of star-kernel decomposition. The square consisting of four faces is the target polygon and normal vectors are given as outward directions. The blue and red regions are star kernels of the decomposed polygons.

is the desired star kernel. For the remaining faces, we iterate this procedure until we reach the last face. The pseudocode is shown in Algorithm 1, and an example of star-kernel decomposition is shown in Fig. 2.

3 EXPERIMENT

To verify the applicability of the obtained placement of multiple projectors obtained by the proposed method, we performed some experiments. We conducted the experiments using a simulation program implemented in Python 3.7 and ran the program on a PC (Intel Core i7-6700, 16GB Memory, Windows 10). We chose two 3D objects as our basic projection targets: a cube and an object consisting of four cubes connected together. We visualized the obtained projector placements and its projection results by placing 3D objects on a workspace of a PC software (Unity 2018.4.30f1) and light sources at the locations obtained by our algorithm.

Fig. 3 shows the obtained placement of multiple projectors, and Fig. 4 shows its simulated projection result. We were able to obtain a projector placement that enabled two projectors to illuminate all faces of the two objects. By placing the light source in the obtained arrangement, it was confirmed through the simulation that it was possible to illuminate all the faces of objects. The execution time of the program to obtain the placement (average of 10-running) was 10.9 ms for the cube and 14.0 ms for the connected four cubes. This result is fast compared to previous heuristic methods, which took time in the order of minutes.

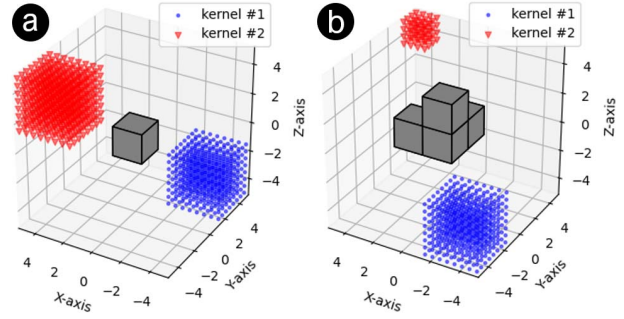


Figure 3: Obtained projector placement of multiple projectors. (a) Result of a cube. (b) Result of the connected four cubes.

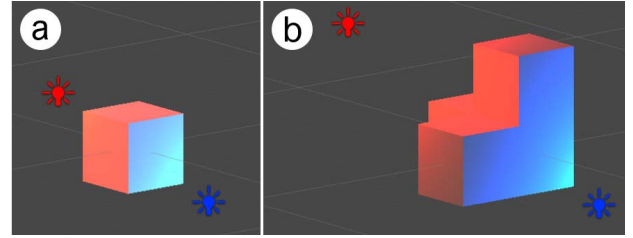


Figure 4: Simulated projection result of the obtained projector placement. (a) Result of a cube. (b) Result of the connected four cubes.

4 CONCLUSION

In this paper, we proposed a novel viewpoint planning method of projector placement using the star-kernel decomposition algorithm of polygons. We introduced the algorithm and implemented a program to obtain candidates for the projector placement. Through simulation experiments on simple 3D objects, it is assumed that our proposed method can rapidly find an appropriate projector placement for illuminating all faces of the objects. For future work, we would approach the projector placement problem including the optimization of luminance and contrast of projection images by adding additional conditions to the search using the proposed method.

ACKNOWLEDGMENTS

This work was supported by JST ACT-X JPMJAX1900, JPMJAX190N, JPMJAX1903, and Osaka City University Advanced Mathematical Institute (MEXT Joint Usage/Research Center on Mathematics and Theoretical Physics JPMXP0619217849).

REFERENCES

- [1] A. J. Law, D. G. Aliaga, and A. Majumder. Projector placement planning for high quality visualizations on real-world colored objects. *IEEE Transactions on Visualization and Computer Graphics*, 16(6):1633–1641, 2010.
- [2] J. M. Lien. Approximate Star-Shaped Decomposition of Point Set Data. In R. P. M. Botsch, ed., *Proceedings of the 2007 Eurographics Symposium on Point-Based Graphics*, pp. 73–80, 2007.
- [3] R. Raskar, G. Welch, K.-L. Low, and D. Bandyopadhyay. Shader Lamps: Animating Real Objects With Image-Based Illumination. In *Proceedings of the 12th Eurographics Workshop on Rendering Techniques*, pp. 89–102, jun 2001.
- [4] W. Yu, M. Li, S. S. Iyengar, and X. Li. Efficient 3D region guarding for multimedia data processing. In *Proceedings of the 2011 IEEE International Conference on Multimedia and Expo - ICME '11*, 2011.